

TeachInVR: A virtual reality classroom for remote education

Florian Schier*

Krishnan Chandran†

Matthew McGinity‡

IXLAB
Technische Universität Dresden

Figure 1: *TeachInVR* - a virtual reality classroom for immersive multi-user teaching and learning experiences

ABSTRACT

TeachInVR is a virtual reality (VR) classroom for remote and co-located group teaching. It supports a range of teaching activities from lectures and student presentations to guided tours, group work and private study. It allows rapid creation of classes through re-use of traditional teaching materials, reducing the barriers for time-constrained teachers. It is also intended as a sandbox environment for exploring teaching concepts that are well supported by, or even unique to, VR. In particular, it provides tools for the spatial arrangement of teaching material and leveraging the spatial properties of VR for learning and recall.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Education; Spatial Association

1 INTRODUCTION

Sharing knowledge and handing it to future generations is, and always has been, an essential part of society. Whilst learning and teaching concepts have long been a focus of research, they are yet not fully understood. There are many well established methods that are used by teachers every day in their classes. TeachInVR aims to be yet another tool teachers have at their disposal. This application does not seek to replace or compete with physical class rooms, but to extent possible ways of passing on said knowledge.

While VR is being successfully used in a wide range of application domains and for a variety of objectives, its potential for teaching is not fully explored, nor fully used. Most VR applications used for teaching are either specific to a field (i.e. "The VR Chemistry

Lab"¹), or even specific to a certain topic within a field (i.e. "Human Anatomy VR"²). These represent bespoke teaching scenarios with long development times for each scenario.

TeachInVR, on the other hand, is designed to be platform that can be used to teach a variety of classes. It therefore does not present an alternative to specific VR learning tools, but rather an extension. By re-imagining various teaching methodologies such as classic 2D slideshows in VR, we strive to provide an environment where teachers can not only teach any subject, but employ a range of teaching styles, from lectures and presentations, to group work and group discussions.

Special attention is given to simplicity of use and the ease and speed with which a class can be prepared. Many teachers have very limited time and resources to prepare each class, and even less time to learn a new tool or style of teaching. Our application, thus, is not only designed for rapid development of classes using existing materials, such as text, images, video, but to support familiar teaching methods. TeachInVR aims to be fully usable with as little training as possible, by purposefully avoiding complicated inputs and hidden menus. This not only lowers entry hurdles, but allows users to focus on user-to-user communication rather than user-system-interactions.

Apart from extending existing teaching methods, this application can be used to connect users remotely, too. Especially during the ongoing COVID-19 pandemic, this may be of help whenever restrictions demand that students study from home. Representing the user by an avatar, immersing them in VR and bringing others into this environment seemingly adds greatly to the perceived social presence that users experience, as our preliminary study hints.

Remote teaching via video chat, while allowing physically distanced lectures and thus safe teaching in the light of the COVID-19 pandemic, brings a number of negative aspects with it. Students are easily distracted, and motivation and interactions are hindered. By utilizing VR, we eliminate some of these distractions, for example smartphones or web-browsing, and present them with - potentially - more engaging experiences. Thus, in future studies, this project will

*e-mail: florian.schier@tu-dresden.de

†e-mail: krishnan.chandran@tu-dresden.de

‡e-mail: matthew.mcginity@tu-dresden.de

¹(<https://www.oculus.com/experiences/quest/3919613214752680>)

²(<https://www.oculus.com/experiences/gear-vr/1658650407494367>)

be compared with video chat lectures and slide shows.

This application enables users to teach, learn and study together in virtual reality. Additionally, the application allows us to leverage and study the benefits of presenting information in a 3D, explorable spatial context. Associating information with salient features of an environment to aid recall has long been utilized. Given limited information storage options, "Memory Palaces", or "method of loci" were developed by the ancient Greek culture. Börner describes them as "a non-linear storage system or random access memory that is responsive to the user's position in an imagined space". [1]

Nowadays this method is utilized in World Memory Championships. At the same time, its potential for individuals without professional memorizing background is actively researched, for example in [4, 12].

Virtual reality allows for the spatial and contextual arrangement and presentation of material that would normally not be embedded in an environment. Images, videos, text, 3D models and sounds are given position, size and spatial relationships with each other and the environment. A lecture, for example, can be presented as a physical journey, a guided walk along a path with not just a beginning and end, but branches, loops, rooms and bridges. Further, these spatial constellations can be persistent, such that students returning to the same virtual classroom week after week can always expect to find the material in its rightful place. It is possible that these spatial presentations will excite the same underlying mnemonic mechanisms at work in the method of loci, and so aid learning and recall.

TeachInVR is also designed as a research tool for exploring the potential of virtual reality-based teaching. We have designed it with the following questions in mind:

- How can teachers and students easily and rapidly use VR for their classes?
- How can we re-use existing teaching materials and know-how?
- What kinds of teaching activities and formats are well supported by VR?
- What kinds of novel teaching formats are only possible with VR?
- How can the inherent spatiality of VR be utilized to aid teaching and learning?

2 RELATED WORK

2.1 Virtual Reality in a teachers toolkit

From the extensive body of research concerning the use of virtual reality for education and learning, we provide here a very brief overview of the work most relevant to this project.

Concerning the efficacy of VR-based education, some prior work finds an improved learning outcome when using VR applications with higher immersion levels (i.e. [6, 10] and partially in [7]). Others, however, find the opposite (i.e. [5]). Comparability is difficult, as studies vary in methods and metrics, teaching concepts and declarative theories. They vary in the content taught (i.e. procedural or declarative knowledge), as well as what they compare their work against. Nevertheless, these studies provide valuable insight on parameters that might improve the success of VR as an educational tool.

In their "systematic review of immersive virtual reality applications for higher education", Radianti et al. provide an overview of work published in 2016 - 2018. Their work reveals aspects that are missing in many applications, most prominently missing interaction with other users, and not allowing user generated content. [8] However, more recent work increasingly includes both.

Most of the applications reviewed by Radianti et al. belong to specific domains. Similar to other authors, such as [2, 11], Radianti et al. emphasize the need for future research to allow for more



Figure 2: A top-down view into the rooms structure. The students are gathered in the center of the classroom.

generalization. Thus, this project provides a platform that can be used to teach a variety of subjects.

There are a number of active research projects striving to create generic platforms for teaching in VR. However, they pursue different goals, as they, for example, build specific interactive scenarios, or do not aim to explore how spatiality may aid learning. Examples are the "Mobility System Cooperation in Higher Education" (*MyScore*) project³ and the *ViLeArn* project⁴.

2.2 Spatial association of information

Mnemonic techniques, such as *Memory Palaces*, aid the recall of information by associating information with salient features of a visualized environment. Prior work suggests that these Memory Palaces cannot exclusively be created mentally, and that VR can be a helpful tool to create them too. Studies show that using VR as a tool for generating Memory Palaces improves memory recall for untrained users. [4, 12]

Besides passive observation of information, work from Jang et al. indicate that interaction with objects aids recall, too. [3]

3 TEACHINVR

TeachInVR allows immersive multi-user teaching and learning experiences that can be conducted remotely and safely. It enables us to move beyond 2D video chat lectures and allows us to bring students to virtual places and present them with complex 3D structures. Architecture students, for example, could present their modelling work in these virtual spaces, rather than in 2D slide shows.

The application is designed to allow multiple users, teachers and students alike, to join a virtual space, and to communicate and interact intuitively with each other and with the environment. TeachInVR offers many natural social and behavioral cues that may be difficult to utilize in video chat. For example, addressing another person by simply facing them while speaking.

Before developing our own sandbox application, we reviewed several commercial platforms available at the time. These evaluations included a group of students meeting weekly on platforms such as Spatial.io, Mozilla Hubs, AltspaceVR, and Engagevr.io. While Spatial.io, at the time, was closest to what we envisioned, essential features were missing, limiting our future research goals. To gain control over the teachers' workflow when creating lectures, to dynamically add features, as well as to manage system behavior, and in order to match privacy concerns, we decided to create TeachInVR as a platform for future research. In turn, we are able to extend this application rapidly. We are, for example, extending it to a hybrid teaching tool, utilizing the Oculus Quests video pass-through to

³(<https://myscore.lfi.rwth-aachen.de>)

⁴(<https://vilearn.hci.uni-wuerzburg.de>)

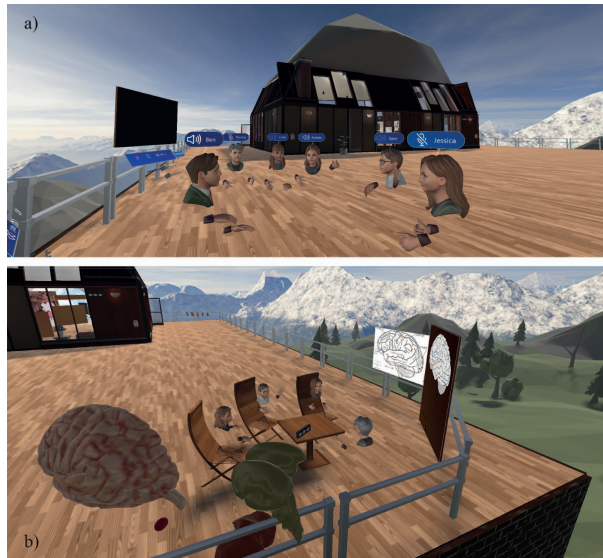


Figure 3: a) Example of a group of students discussing a group task. b) Discussion on the outside patio. The large open space allows the teacher to separate student groups and still observe all of them.

teach collocated and remote students concurrently. Similarly, with this platform, we can swiftly address teachers' and students' wishes and requirements, gathered in our trials with real classes.

Focus on the teachers requirements

Given the limited time of teachers to prepare classes, let alone getting to know yet another software, this application allows them to set up their lessons without having to understand the applications inner workings, nor requiring them to be proficient in 3D software development.

Creating a lecture in TeachInVR only requires a few steps. Teachers can bring videos, images, audio and 3D models into the environment and arrange them spatially from within VR. The system ensures that every connected participant experiences the same virtual world. Videos and states are efficiently kept in sync to avoid diverging experiences and to keep every participants focus on the content of the lecture, rather than technical difficulties.

Teachers then use the space to steer the appearance and sequence of material, either by moving assets in and out of the conversation, or by physically guiding students on a journey through the material.

3.1 TeachInVR: Features & Application Design

3.1.1 Avatar

Recent work by Ripka et al. emphasizes, that missing social and behavioral cues can make group interactions more challenging. Yet, their participants still reported a strong sense of co-presence. [9] Based on these findings, this application uses basic cues, such as volume controlled mouth movements and randomized eye movement. It also utilizes the Oculus Quests powerful hand tracking to increase intuitiveness and social perception. It is assumed that further improvements, for example in capturing the users facial expressions, are under development by the hardware developers.

The application features a customizable appearance build upon the ReadyPlayerMe platform⁵. These avatars are synced across all devices, and stored locally to reduce workload when users rejoin. Both, hands and controllers are supported, allowing the user to choose their preferred interaction method. Furthermore, spatialized

⁵(<https://readyplayer.me>)

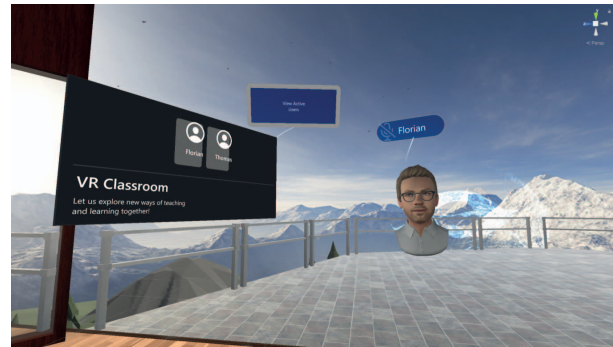


Figure 4: Exemplary view of a muted user and the Active Users board.

voice transmission aids the natural communication with other users. Additionally, each avatar features an indicator, displaying their name and indicating their voice volume. For better readability, these always face the local user (see Fig. 3.a and Fig. 4).

3.1.2 Virtual Content

The application supports different types of media - texts, images, 3D models, videos and audio. To ensure all participants have the same experience, the system syncs the state, timestamp and volume of video and audio files efficiently. Additionally, the session can be saved and restored at any time, providing the users an extra layer of convenience. This allows them, for example, to restore the initial state at the end of a class and to summarize the lectures content.

3.1.3 Interactions

The system attempts to avoid complicated input gestures to be as intuitive and inclusive as possible. By design, there are no hidden or strongly hierarchical menus and the key features are kept easily accessible. The application provides further convenience features, such as an "active users" board (see Fig. 4) and a non-intrusive notification when the user is muted, offering low affordance to enable the microphone again (see Fig. 1).

Generally, users are able to interact with any imported content. These assets can be moved around, scaled and played back. Thus, content can be put in the center of the conversation, viewed in large scales and put aside again quickly and intuitively.

3.2 Equipment

While the market of consumer level VR hardware evolves, the Oculus Quest continues to be one of the most affordable devices available. Thus, we target this device, allowing classes to be equipped with a sufficient number of devices. The application is developed in the game engine *Unity*⁶ and uses the *Mixed Reality Toolkit*⁷.

4 PRELIMINARY STUDY

The application was evaluated in a preliminary study. An extensive presentation was given in the virtual environment. The study primarily assessed the applications usability and potential for future research and large scale studies.

A wide range of different teaching methods for teaching in VR are possible. For this study, we decided on a circular arrangement of the lectures content. Thus, the content surrounded the class and the presenter guided the participants clockwise through the lectures material (see Fig. 2). It was hypothesized that this layout should aid the users learning by allowing spatial associations between the material and salient features of the environment.

⁶(<https://unity.com>)

⁷(<https://docs.microsoft.com/en-us/windows/mixed-reality>)



Figure 5: a) Users discussing different assets within the environment. b) Live-view of the user in the center (in a)), selecting where to move.

The presenter utilized 3D models, videos and images, and moved them around to bring them in and out of the conversation while guiding the class spatially through the content.

4.1 Procedure

5 participants (3 female, 2 male) and one lecturer were using the application for approximately 70 minutes. During the first 20 minutes, participants were allowed to set up their avatars and get familiar with the system. Subsequently, the lecture on "The Neurobiological Correlates of Compassion vs. Empathy" ran for about 30 minutes. None of the participants in *student roles* had prior expertise in this area. Students were then divided into groups of two, and asked to analyze video material in regard to the lectures content. The lecture closed with the students presenting their findings to the others. The exercise took about 20 minutes in total. As, in the light of the ongoing COVID-19 pandemic, this preliminary study was run remotely, the participants were asked to complete a survey on their computers.

Besides common questionnaires regarding the systems usability, perceived and social presence, the survey also aimed to compare the presented system to video chat lectures. It was assumed, given the COVID-19 pandemic, that participants had adequate prior experience participating remote video chat lectures.

4.2 Results

This preliminary study confirmed that the environment was fully usable with little to no training. A short video introducing the available interactions was sufficient for all participants to grasp the systems behaviour and interface. Furthermore, the system proved to be robust and reliable, as the participants reported no technical difficulties and the teacher was able to fully focus on presenting their lecture and managing the class.

Compared with video chat lectures, most participants reported an increased sense of co-presence and "having met others". Some especially liked the "possibility to interact with every piece of visualized information, [making] the lesson more interesting and exciting".

Additionally, while not being in the scope of this survey, one participant highlighted that they were able to recall information easier than being presented with slides. They argued that visualizing the room recalling the information helped them.

5 CONCLUSION AND FUTURE WORK

TeachInVR provides a virtual reality space that enables an immersive multi-user teaching and learning experience. The application can be used in a typical classroom setting, but can also be used to connect students remotely. Enabling teachers to rapidly bring their existing content into VR and explore new teaching methods without requiring software development skills lowers entry hurdles and will encourage adaptation. The application supports students and teachers to teach, present, study and collaborate, and provides them with a high sense of co-presence and engagement. The environment maintains persistent spaces to allow students to revisit lectures, prepare presentations, or to work by themselves or in groups.

At the same time, TeachInVR provides a sandbox environment to rapidly implement, test and evaluate methods of teaching in VR. It will be used to further analyze VRs potential as a helpful tool for teachers and help to understand in which situations and ways VR can be useful for teaching. As previously described, many researchers wish for future research to allow for more generalization, and we hope to contribute to the ongoing progress in this field - providing not a bespoke application to teach a specific topic, but instead an additional, generic tool teachers can easily use in their classes, regardless of their subject.

This project also lies the foundation for future studies, assessing the potential benefits of spatial association of information and their use in the context of both, teaching and learning. Using TeachInVR, we are beginning a series of trials with real classes. Our goal is to evaluate experiences and requirements of teachers and students.

A brief video demonstrating core concepts of TeachInVR can be found here.⁸

ACKNOWLEDGMENTS

The authors wish to thank all participants that participated in our evaluations, as well as Lynda Joy Gerry for giving the said lecture.

REFERENCES

- [1] K. Börner. iScape: A collaborative memory palace for digital library search results. p. 5.
- [2] J. Harman, R. Brown, and D. Johnson. Improved memory elicitation in virtual reality: New experimental results and insights. In R. Bernhaupt, G. Dalvi, A. Joshi, D. K. Balkrishnan, J. O'Neill, and M. Winckler, eds., *Human-Computer Interaction - INTERACT 2017*, vol. 10514, pp. 128–146. Springer International Publishing. Series Title: Lecture Notes in Computer Science. doi: 10.1007/978-3-319-67684-5_9
- [3] S. Jang, J. M. Vitale, R. W. Jyung, and J. B. Black. Direct manipulation is better than passive viewing for learning anatomy in a three-dimensional virtual reality environment. 106:150–165. doi: 10.1016/j.compedu.2016.12.009
- [4] E. Krokos, C. Plaisant, and A. Varshney. Virtual memory palaces: immersion aids recall. 23(1):1–15. doi: 10.1007/s10055-018-0346-3
- [5] G. Makransky, T. S. Terkildsen, and R. E. Mayer. Adding immersive virtual reality to a science lab simulation causes more presence but less learning. 60:225–236. doi: 10.1016/j.learninstruc.2017.12.007
- [6] D. M. Markowitz, R. Laha, B. P. Perone, R. D. Pea, and J. N. Bailenson. Immersive virtual reality field trips facilitate learning about climate change. 9:2364. doi: 10.3389/fpsyg.2018.02364
- [7] K. A. Pollard, A. H. Oiknine, B. T. Files, A. M. Sinatra, D. Patton, M. Ericson, J. Thomas, and P. Khooshabeh. Level of immersion affects spatial learning in virtual environments: results of a three-condition within-subjects study with long intersession intervals. 24(4):783–796. doi: 10.1007/s10055-019-00411-y
- [8] J. Radiant, T. A. Majchrzak, J. Fromm, and I. Wohlgenannt. A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. 147:103778. doi: 10.1016/j.compedu.2019.103778
- [9] G. Ripka, S. Grafe, and M. E. Latoschik. Preservice teachers' encounter with social VR – exploring virtual teaching and learning processes in initial teacher education. In E. Langran, ed., *Proceedings of SITE Interactive Conference 2020*, pp. 549–562. Association for the Advancement of Computing in Education (AACE).
- [10] R. Webster. Declarative knowledge acquisition in immersive virtual learning environments. 24(6):1319–1333. doi: 10.1080/10494820.2014.994533
- [11] R. Webster and J. Dues. System usability scale (SUS): Oculus rift® DK2 and samsung gear VR®. In *2017 ASEE Annual Conference & Exposition Proceedings*, p. 28899. ASEE Conferences. doi: 10.18260/1-2-28899
- [12] F. Yang, J. Qian, J. Novotny, D. Badre, C. Jackson, and D. Laidlaw. A virtual reality memory palace variant aids knowledge retrieval from scholarly articles. pp. 1–1. doi: 10.1109/TVCG.2020.3009003

⁸(<https://vimeo.com/666164862>)